

Assignment 9

Q1

Let X be the set $\{1, 2, 3, 4, 5\}$ and Y be the set $\{6, 7, 8, 9, 10\}$. We describe the functions $f: X \rightarrow Y$ and $g: X \rightarrow Y$ in the following tables. Answer each part and give a reason for each negative answer.

n	$f(n)$
1	6
2	7
3	6
4	7
5	6

n	$g(n)$
1	10
2	9
3	8
4	7
5	6

- Is f injective?
- Is f surjective?
- Is f bijective?
- Is g injective?
- Is g surjective?
- Is g bijective?

A1

a. No b. No c. No d. Yes e. Yes f. Yes

Q2

Let T be the set of all infinite sequences over $\{0,1,2\}$. Show that T is uncountable using a proof by diagonalisation.

A2

Proof:

Assume T is countable, so there is a 1-1 correspondence $f: \mathbb{N} \rightarrow T$.

n	$f(n)$
1	00000...
2	11111...
3	22222...
4	01201...
5	21021...
...	...

Demonstrate an infinite sequence $x \in T$ that is missing from the list.

$x = 12012...$ differs from the n^{th} number in the n^{th} digit, so that x cannot be the n^{th} number for any n .

Hence x is not paired with any n . It is missing from the list.

Therefore f is not a 1-1 correspondence, which is a contradiction.

Hence, T is uncountable.

Q3

Let $O = \{(i, j, k) \mid i, j, k \in \mathbb{N}\}$. Show that O is countable.

A3

Proof:

Itemize the tuples of $\{(i, j, k) \mid i, j, k \in \mathbb{N}\}$ in the following order.

n	Tuples
1	(0, 0, 0)
2	(0, 0, 1)
3	(0, 1, 0)
4	(1, 0, 0)
5	(0, 0, 2)
6	(0, 1, 1)
7	(0, 2, 0)
8	(1, 0, 1)
9	(1, 1, 0)
10	(2, 0, 0)
11	(0, 0, 3)
$\vdots$	$\vdots$

To be specific, we list the tuples whose sum of all three elements are lower first. For the tuples whose sum are the same, we arrange them in increasing order of $(i + j + k)^2 \cdot i + (i + j + k) \cdot j + k$ for (i, j, k) .

In this case, the transformation between $\mathbb{N}$ and O is obviously both injective, for different n , the corresponding element in O is different, and surjective, every element of O can be found in the list. So it is bijective and therefore O is countable.

Q4

Prove that it is decidable to determine whether a Turing machine ever writes a non-blank symbol when started on a blank tape.

A4

Proof:

Let

$$W = \{\langle M \rangle \mid M \text{ ever writes a non-blank symbol when started on a blank tape}\}.$$

We construct a TM T deciding W .

$T =$ "On input $\langle M \rangle$:

1. Read all the states of M . Let n be the number of states of M .
2. Simulate M on a blank tape.
3. If during the simulation M writes a non-blank symbol, accept.
4. If M enters some state for the second time before writing any non-blank symbol, reject.
5. If M halts before writing any non-blank symbol, reject."

This algorithm always halts. Indeed, before M writes any non-blank symbol, the tape remains entirely blank. Hence the behavior of M only depends on its current state. If some state repeats before any non-blank symbol is written, then M will repeat the same sequence of states and actions forever, and so it will never write a non-blank symbol.

Since M has only finitely many states, if no non-blank symbol is written within the first n transitions, some state must repeat. Therefore T halts on all inputs and correctly decides W .

Hence W is decidable.

Q5

Is the question "Does a Turing machine have at least 481 states" decidable? Explain your answer.

A5

Yes, it is decidable.

We can build a TM M reading the input TM $\langle M' \rangle$ and count the states in Q of the formal description, or say encoding, of M' , until the number reaches 481 or there are no more states.

$M =$ "On input $\langle M' \rangle$:

1. Set a counter and start from 0.
2. Count the states in the state set of M' . Add 1 to the counter for every new read state. If the counter reaches 481, accept. Otherwise, if there are no more states and the counter is smaller than 481, reject."

Or we can build a TM M reading the input TM $\langle M' \rangle$ and count all the states in Q of the formal description of M' . Then check whether the number is bigger than or equal to 481. This process must halt since the number of states of M' is finite.

Q6

Let $A = \{ \langle M \rangle \mid M \text{ is a DFA that doesn't accept any string containing an even number of 1s} \}$. Show that A is decidable.

A6

Let

$$A = \{ \langle M \rangle \mid M \text{ is a DFA and } L(M) \cap (0^*10^*10^*)^* = \emptyset \}$$

Give TM D_A that decides A ,

$D_A =$ "on input $\langle M \rangle$

1. Construct a DFA P such that $L(P) = (0^*10^*10^*)^*$,
2. Construct a DFA Q such that $L(Q) = L(P) \cap L(M)$,
3. Run TM E_{DFA} on input $\langle Q \rangle$,
4. Accept if E_{DFA} accepts; otherwise, reject."

All steps are computable and finite, so A is decidable.